

Introduction to Microeconomics

Lecture 12—Income and Substitution Effects & Expenditure Minimization

Prof. Alejandro Melo-Ponce



NAZARBAYEV
UNIVERSITY

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Income and Substitution Effects

A fall in price of a good has two effects:

- ① Consumers will tend to buy more of the good that has become cheaper and less of those goods that are now relatively more expensive. This response to a change in the relative prices of goods is called the **substitution effect**.
- ② Because one of the goods is now cheaper, consumers enjoy an increase in real purchasing power. The change in demand resulting from this change in real purchasing power is called the **income effect**.

Examples: an almost pure income effect and an almost pure substitution effect

- Suppose we are considering the rent that needs to be paid for living in an apartment versus other goods.
 - Suppose there's a decrease in the rental price of your unit.
 - Rent involves a **large share of your income**.
 - You will end up feeling “richer” because you will have more disposable income to buy other stuff.
 - This is an example of an (almost pure) income effect.
- What apple to buy? Granny Smith apples (green skin, crisp and juicy flesh, tart and acidic flavor) or McIntosh apples (red/green skin, tender white skin, tart flavor).
 - This purchase involves a very little share of your income.
 - Suppose that you just want an apple and thus, you are equally happy between both varieties.
 - You will end up buying the one that is cheaper.
 - If prices change, you buy the one with the lowest price, that is, you **substitute towards the relatively cheaper alternative**.
 - This is an example of an (almost pure) substitution effect.

Definitions

- The **substitution effect** is the change in consumption of a good associated with a change in its price, with the level of utility held constant.
- The **income effect** is the change in consumption of a good resulting from an increase in purchasing power, with relative prices held constant.

The Total effect of a price change and its decomposition

The total effect after a price change

- Suppose that we want to analyze the change in quantity demanded of good 1 after a price change from p_1^0 to p_1^1 , while holding the price of good 2 and income fixed. Then the **Total effect of the price change**, or TE , is

$$TE = x_1^m(p_1^1, p_2, I) - x_1^m(p_1^0, p_2, I)$$

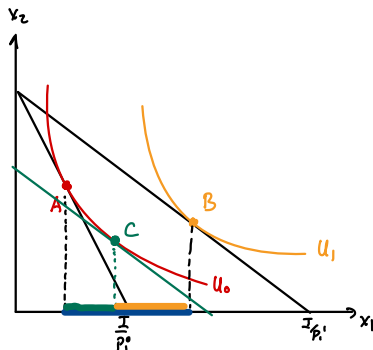
or for simplicity:

$$TE = x_1(p_1^1) - x_1(p_1^0).$$

- The goal is to decompose the TE into the substitution effect SE and the income effect IE .
 - The SE will respect the law of demand (goes in opposite direction to the price change).
 - The IE depends on whether the good is normal or inferior.

Graphical Analysis 1A: Normal good and a price decrease

$$p_1^0 > p_1^1 \Rightarrow \Delta p_1 = p_1^1 - p_1^0 < 0, \quad p_2, I \text{ fixed}$$



$$A = (x_1^m(p_1^0, p_2, I), x_2^m(p_1^0, p_2, I))$$

$$B = (x_1^m(p_1^1, p_2, I), x_2^m(p_1^1, p_2, I))$$

$$C = (x_1^m(p_1^1, p_2, I^c), x_2^m(p_1^1, p_2, I^c))$$

$$\underline{TE = x_1^m(p_1^1) - x_1^m(p_1^0)}$$

To decompose the TE into SE and IE we need to take money away from the consumer to leave her at the original level of utility U_0

The green budget line $I^c < I, \Delta I = I - I^c > 0$

$$\underline{SE = x_1^m(p_1^1, p_2, I^c) - x_1^m(p_1^0, p_2, I) > 0}$$

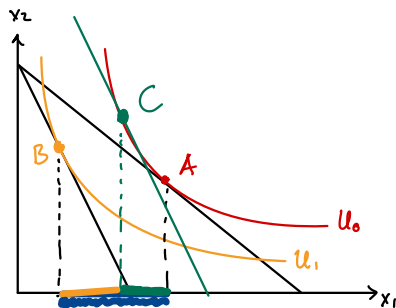
$$\underline{IE = x_1^m(p_1^1, p_2, I) - x_1^m(p_1^1, p_2, I^c) > 0}$$

$$\begin{aligned} TE &= SE + IE \\ &= x_1^m(p_1^1, p_2, I) - x_1^m(p_1^0, p_2, I) > 0 \end{aligned}$$

For a normal good, the TE will always satisfy the Law of Demand. SE & IE have the same sign!

Graphical Analysis 1B: Normal good and a price increase

$p_1^0 < p_1^1 \Rightarrow \Delta p_1 = p_1^1 - p_1^0 > 0$, p_2, I are fixed.



$$A = (x_1^m(p_1^1, p_2, I), x_2^m(p_1^1, p_2, I))$$

$$B = (x_1^m(p_1^0, p_2, I), x_2^m(p_1^0, p_2, I))$$

$$TE = x_1^m(p_1^1, p_2, I) - x_1^m(p_1^0, p_2, I)$$

To decompose TE , we need to leave the consumer at the utility level U_0 with the new prices

The green line has income $I^c > I$, thus we need to give money to the consumer:

$$\Delta I = I^c - I$$

$$SE = x_1^m(p_1^1, p_2, I^c) - x_1^m(p_1^0, p_2, I) < 0$$

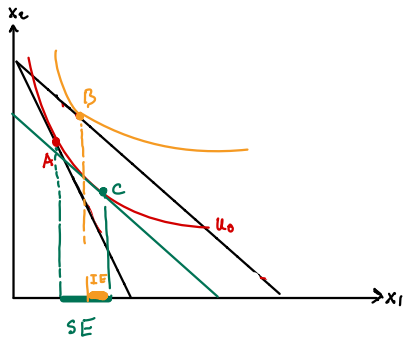
$$IE = x_1^m(p_1^1, p_2, I) - x_1^m(p_1^1, p_2, I^c) < 0$$

$$\underline{TE} = x_1^m(p_1^1, p_2, I) - x_1^m(p_1^0, p_2, I) < 0$$

SE & IE go in the same direction for a normal good.

Graphical Analysis 2A: Inferior good and a price decrease

$$p_1^0 > p_1^1 \Rightarrow \Delta p_1 = p_1^1 - p_1^0 > 0, \quad p_2, I \text{ fixed}$$



$$TE = X_1(p_1^1, p_2, I) - X_1(p_1^0, p_2, I) > 0$$

$$SE = X_1(p_1^1, p_2, I^c) - X_1(p_1^0, p_2, I) > 0$$

$$IE = X_1(p_1^1, p_2, I) - X_1(p_1^1, p_2, I^c) < 0$$

$\hookrightarrow$ inferior good

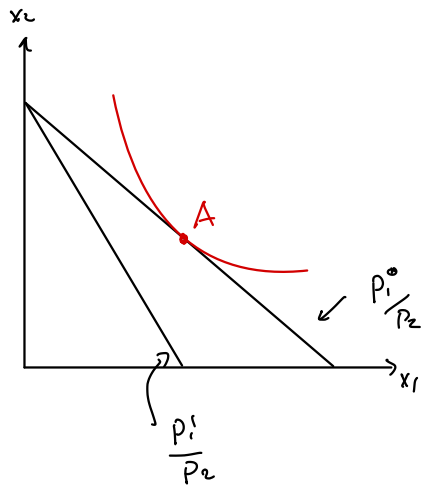
$$|SE| > |IE| \Rightarrow TE > 0$$

SE dominates the income effect $\Rightarrow TE$ satisfies the law of demand.

Graphical Analysis 2B: Inferior good and a price increase (homework!)

$p_1' > p_1^0$, p_2, I constant

Assume $|SE| > |IE|$



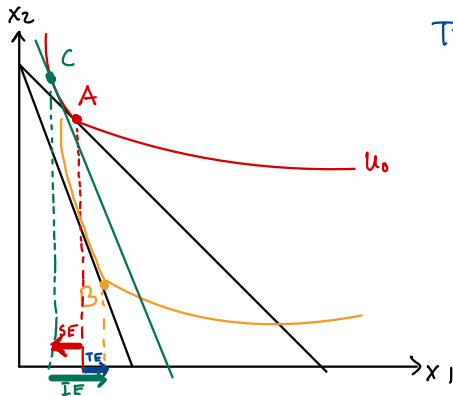
A theoretical Paradox: Giffen Goods and upward sloping demand.

This case can only happen for inferior goods, for which $|SE| < |IE|$

price increase $p_i' > p_i^0$

SE goes in opposite direction to the IE

TE = movement from A to B (on the x_i axis) > 0



↳ Can happen with poor households and staple goods.

The expenditure minimization problem

In order to be able to calculate analytically the decomposition of the total effect into the income and substitution effects, we will need to analyze “dual problem” to the utility maximization problem of the consumer. Recall:

$$\max_{x_1, x_2} u(x_1, x_2) \quad \text{subject to } p_1 x_1 + p_2 x_2 \leq I \quad (\text{UMP})$$

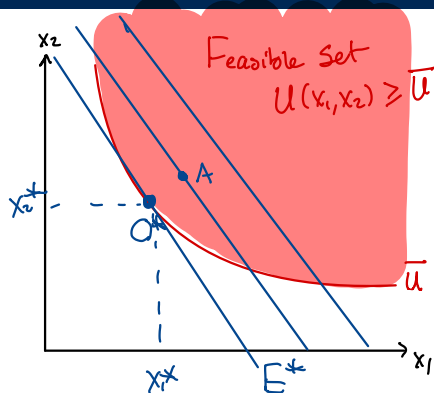
Expenditure Minimization

Given preferences, prices and a fixed level of utility $\bar{U}$, find the cheapest consumption bundle that achieves a level of utility of at least $\bar{U}$.

$$\min_{x_1, x_2} p_1 x_1 + p_2 x_2 \quad \text{subject to } u(x_1, x_2) \geq \bar{U} \quad (\text{EMP})$$

The parameters of this problem are p_1, p_2 and the utility level $\bar{U}$. The solutions to this problem, when prices change, will give an optimal bundle that achieves exactly the original utility level, but with a different price ratio.

Graphical Analysis of the EMP and its optimality condition



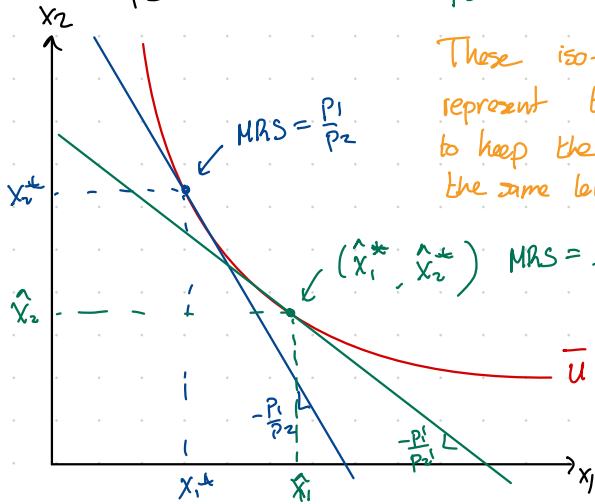
Expenditure $p_1 x_1 + p_2 x_2 = E$
iso-expenditure line

$$U(A) > \bar{u}$$

$$E^* = p_1 x_1^* + p_2 x_2^*$$

$$\text{Optimality Condition: } MRS = \frac{p_1}{p_2}$$

What happens when $\frac{P_1}{P_2}$ changes? $\rightarrow \frac{P_1'}{P_2'}$



These iso-expenditure lines represent the income required to keep the consumer at the same level of utility $\bar{U}$ but with different price ratios.

Steps to decompose the total effect into the income and substitution effects

Suppose that we hold income fixed at I and the price of good 2 at p_2 . Let the initial price of good 1 be p_1^0 which then changes to p_1^1 .

- 1 Compute the demand functions $x_1(p_1^0, p_2, I)$ and $x_2(p_1^0, p_2, I)$.
- 2 Compute the achieved level of utility at those demands: $\bar{U} = u(x_1(p_1^0, p_2, I), x_2(p_1^0, p_2, I))$.
- 3 Compute the new demands when the price of good one changes to p_1^1 : $x_1(p_1^1, p_2, I)$ and $x_2(p_1^1, p_2, I)$.
- 4 The total effect is $TE = x_1(p_1^1, p_2, I) - x_1(p_1^0, p_2, I)$
- 5 To find the income required to leave the consumer at the utility level $\bar{U}$ at the new prices, solve the expenditure minimization problem:

$$\min_{x_1, x_2} p_1^1 x_1 + p_2 x_2 \quad s.t. \quad u(x_1, x_2) = \bar{U}$$

- 6 The solution to the problem above are called compensated demands $x_1^c(p_1^1, p_2, \bar{U})$ and $x_2^c(p_1^1, p_2, \bar{U})$.
- 7 Compute the compensated level of income $I^c = p_1^1 x_1^c + p_2 x_2^c$.
- 8 Note that $x_1^m(p_1^1, p_2, I^c) = x_1^c(p_1^1, p_2, \bar{U})$.
- 9 The substitution effect is $SE = x_1^c(p_1^1, p_2, \bar{U}) - x_1^m(p_1^0, p_2, I)$.
- 10 The income effect is $IE = x_1^m(p_1^1, p_2, I) - x_1^c(p_1^1, p_2, \bar{U})$.

Example with Cobb-Douglas Utility $U(x_1, x_2) = x_1^\alpha x_2^{1-\alpha}$ $0 < \alpha < 1$

$$MRS = \frac{\alpha}{1-\alpha} \frac{x_2}{x_1}, \quad \text{Recall} \quad x_1^m = \frac{\alpha I}{p_1}, \quad x_2^m = \frac{(1-\alpha)I}{p_2}$$

Ex. $\alpha = 0.3$, $p_2 = 1$, $I = 100$, $p_1^0 = 2$, $p_1^1 = 1$, $\Delta p_1 = p_1^1 - p_1^0 = -1 < 0$

$$x_1^m(2, 1, 100) = \frac{0.3(100)}{2} = 15 \quad x_1^m(1, 1, 100) = \frac{0.3 \cdot 100}{1} = 30$$

$$x_2^m(2, 1, 100) = \frac{0.7(100)}{1} = 70 \quad x_2^m(1, 1, 100) = 70$$

$$A = (15, 70), \quad B = (30, 70)$$

$$\bar{U} = U(A) = U(15, 70) = 15^{0.3} \cdot 70^{0.7} \approx 44.0957$$

How do we find I^c ? $\min_{x_1, x_2} x_1 + x_2 \quad \text{s.t.} \quad x_1^\alpha x_2^{1-\alpha} = 44.0957$

Optimality Condition: $\frac{0.3}{0.7} \frac{x_2}{x_1} = \frac{1}{1} \Rightarrow \frac{3}{7} \frac{x_2}{x_1} = 1 \Rightarrow x_2 = \frac{7}{3} x_1$ solve together with

$$\begin{aligned}
 & x_1^{0.3} x_2^{0.7} = 44.0957 \\
 & x_2 = \frac{7}{3} x_1 \quad \text{solve together with} \quad \Rightarrow x_1^{0.3} \left(\frac{7}{3} x_1 \right)^{0.7} = 44.0957 \\
 & \Rightarrow x_1 \left(\frac{7}{3} \right)^{0.7} = 44.0957
 \end{aligned}$$

$$C \begin{cases} x_1^c = \frac{44.0957}{(7/3)^{0.7}} \approx 24.3676 \\ x_2^c = \frac{7}{3} (24.3676) \approx 56.8577 \end{cases}$$

$$I^c = (1)(24.3676) + (1)(56.8577) \approx 81.2253$$

$$\Delta I = I - I^c = 100 - 81.2253 = 18.7747$$

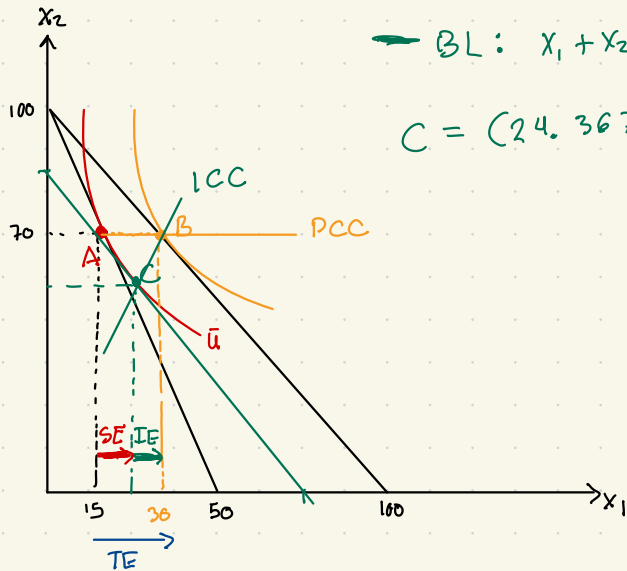
$$x_1^m(1, 1, 81.2253) = \frac{0.3(81.2253)}{1} = 24.3676$$

$$x_2^m(1, 1, 81.2253) = \frac{0.7(81.2253)}{1} = 56.8577$$

$$SE = C - A = 24.3676 - 15 = 9.3676$$

$$IE = B - C = 30 - 24.3676 = 5.6324$$

$$TE = B - A = 30 - 15 = 15$$



$$\text{BL: } x_1 + x_2 = I^c \quad I^c = 81.2253$$

$$C = (24.3676, 56.8577)$$